

EXPERIMENT 3-4. DROOPING ANTENNA

In this experiment, let's the radial pattern of E-Plane and H-Plane for a drooping antenna used for 500MHz. We will study the polarization characteristic of a drooping antenna. Let's calculate the half-power beam width of a drooping antenna by using an antenna software.

BASIC STUDY

1. Drooping Antenna

There is an element of $\lambda/4$ length vertically from a central lead of a coaxial cable as Figure 4-1. Also, as this is the antenna attached a ground lead of $\lambda/4$ length radially in sheath conductor, the lead has an optional angle with earth, this is the antenna attached the counter poise ground lead of a kind instead of a ground of $\lambda/4$ ground antenna.

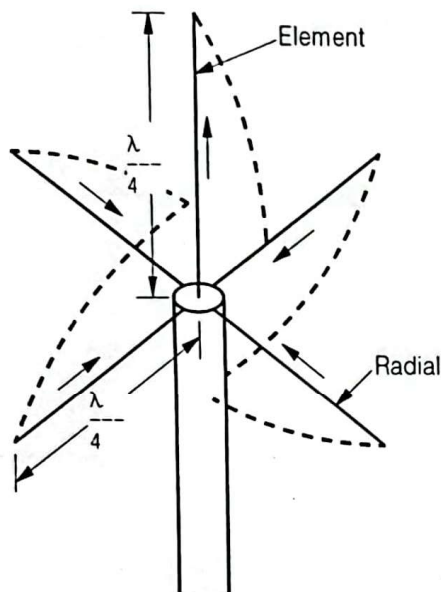


Figure 4-1 Ground-Plane

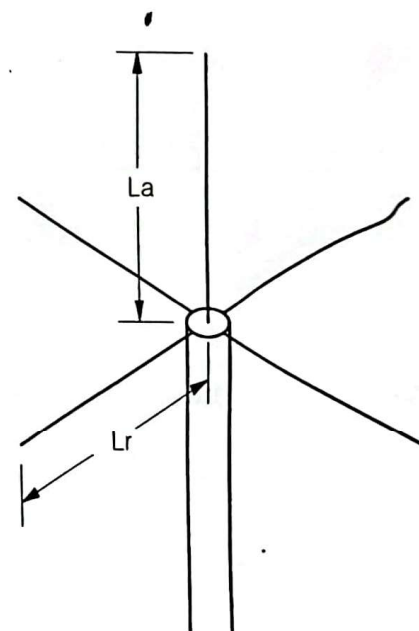


Figure 4-2 Vertical Element L_a

For there is this counter poise ground lead, this is prevented a current from leaking from a sheath conductor of a coaxial cable to external, and there is no unnecessary radiation of an electric wave in cable. Assemble vectorially the radial current placed in optional angle for earth, since the radial length is $\lambda/4$, a radial field is decided by the current flowing into vertical element and the sum of a vector of a radial current.

This antenna has a similar characteristic to $\lambda/2$ dipole antenna, but the radiation of an electric wave can't be occurred almost to the direction of earth, and the radiation is occurred in the area bounded by the central lead and ground lead. Accordingly this is used mainly in a simultaneous call with the airplane taken off and landed in the control tower of airport. The length of a central element is taken by follow formula.(Figure 4-2)

$$L_a = \frac{71.3}{f(\text{MHz})} (m)$$

Also the length of radial is taken by follow formula since make the length of radial to be longer as 2~3% than a vertical element for the radial of an electric wave becomes maximum in vertical direction.

$$L_r = \frac{73}{f(\text{MHz})} (m)$$

Since the impedance at the power feed point of this drooping antenna is very low as $21 \sim 24\Omega$, if feed directly with $50 \sim 75\Omega$ of a coaxial cable, SWR is to be worse. So to go up the impedance at the power feed point, of this antenna, following method has been used.

Figure 4-3 shows the folding shape of a vertical element. Since the impedance at the power feed point becomes 4times of it before fold, can go up to $84 \sim 96\Omega$.

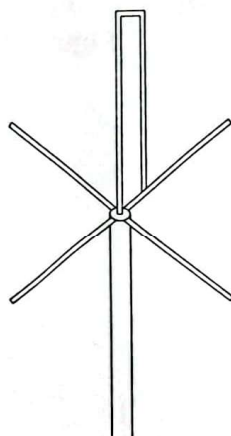


Figure 4-3 Fold the Vertical Element

In figure 4-4, as incline the counter poise ground element to below, the impedance at a power feed point becomes higher.

Figure 4-5 indicates the angle of a ground element and the value of an impedance. this becomes about 21Ω when becomes 90° . and 50Ω in about 3

0 ~ 38 Ω , and becomes 70 Ω when go down just below 0.

Figure 4-6 shows the setting of gamma matching system in a power feed part, Make the impedance at a power feed point to match to the characteristic impedance of a coaxial cable by controlling the length of a gamma-load place parallel with a vertical element, and the variable condenser between the load and the central conductors of a coaxial cable.

Figure 4-7 shows the antenna taken a matching by placing a vertical radial in a coaxial antenna. As connect a parallel radial, d , a distance between a central lead of a coaxial cable and a vertical element, is taken by below formula.

$$d(m) = \frac{\lambda}{2\pi} \cos^{-1} \sqrt{\frac{Z_A}{Z_0}}$$

Z_A : Impedance at a power feed point of drooping antenna

Z_0 : Characteristic Impedance of Coaxial Cable

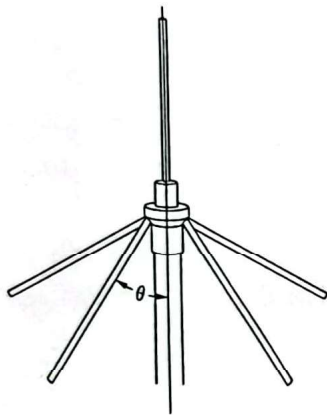


Figure 4-4 Vertical and Impedance

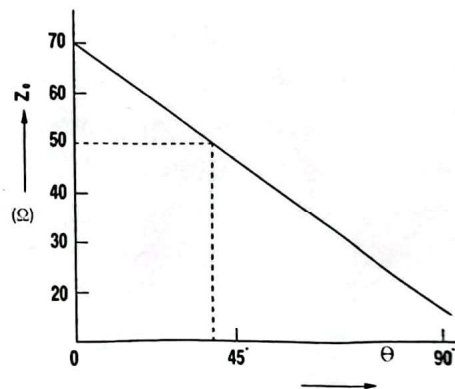


Figure 4-5 Angle θ and Impedance of Vertical Element

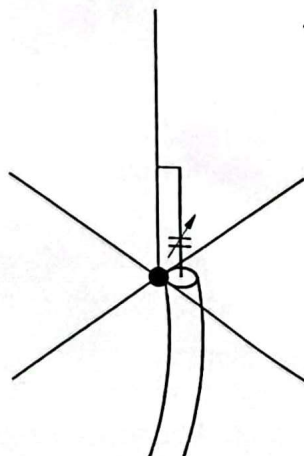


Figure 4-6 Gamma-Match

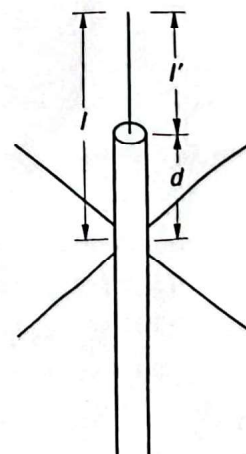


Figure 4-7 Place the Vertical Radial in a Coax

The experiment set of an antenna contain a drooping monopole antenna in addition to a standard monopole. The drooping monopole is very easy design method of monopole. Figure 4-8 shows a drooping monopole exemplarily.

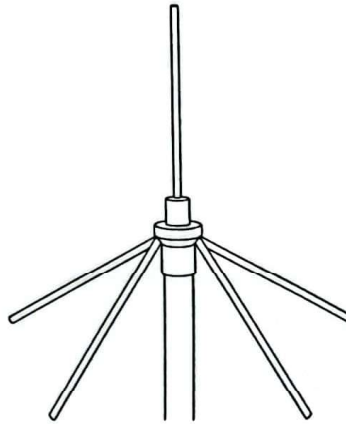


Figure 4-8 Drooping Monopole

To explain the characteristic of a drooping antenna, consider a half-wave dipole having impedance of 73Ω as Figure 4-9. If each leads consisting the half of below part is divided to 180° as Figure 4-9(c), it becomes a $\lambda/4$ ground antenna taking the characteristic impedance of 37.5Ω . But as Figure 4-9(b), if consist of each leads stretching for 90° (45° from a right angle), it becomes a drooping antenna getting a characteristic impedance of almost 50Ω . This is matched exactly up to the characteristic impedance of a coaxial cable.

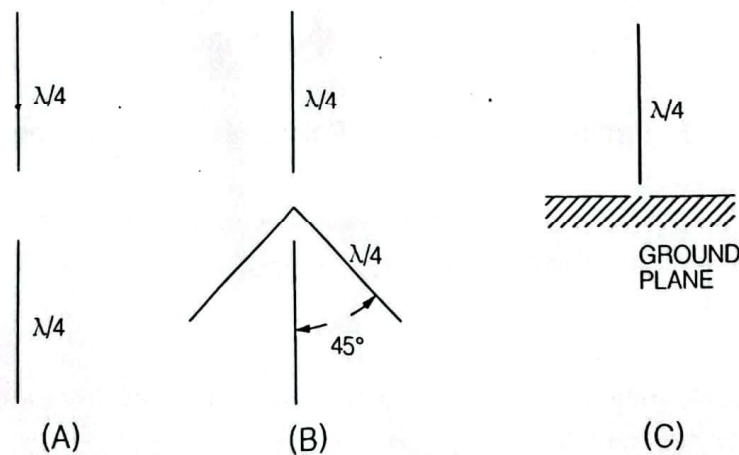


Figure 4-9 (A) Half-Wave Dipole (B) Drooping Antenna (C) $\lambda/4$ Ground Antenna

EXPERIMENT PROCEDURE

SETTING OF EXPERIMENTAL INSTRUMENT

1. Set a transmitting-receiving antenna positioner and a computer which are main instrument of an antenna experimental set.
2. Set a height controller on a transmitter and fix the 500MHz Yagi antenna on a fixing pack of an antenna, then set on the polarization control plate. Set this at a parallel position with earth for the antenna get a parallel polarization characteristic. The transmitting antenna set as Figure 4-10 has a parallel polarization characteristic.

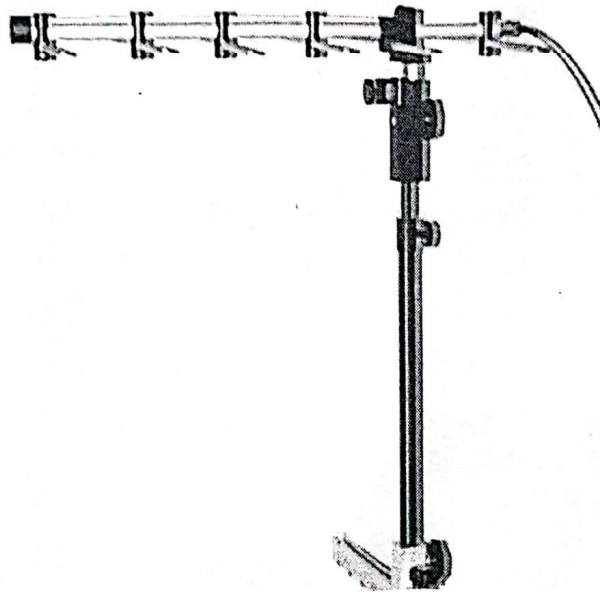


Figure 4-10 Setting of a Parallel Polarization Transmitting Antenna

Set OFF the selection switch and modulation switch in a transmitting of main controller, connect a 500MHz oscillation output terminal and Yagi antenna with SMA used for 2m.

3. Attach the stationary bar on a receiver. And fix the drooping antenna on a polarization control plate of upper stationary bar to be got a parallel polarization characteristic. Set it in center of a rotational axis of a receiver by loosening the position control screw of a receiver and moving the control plate. See Figure 4-11 to set.



Figure 4-11 Setting of Parallel Polarization Receiving Antenna

Connect a coupling terminal of a drooping antenna and a RF IN terminal of a receiver with a SMA cable used for 1m, and connect a OUT PUT(1kHz) terminal of a receiver and a Input(1kHz) terminal of a main controller with a BNC cable used for 1m.

4. Calculate the length of a drooping antenna and measure the actual length of an antenna in 500MHz by using below formula, and record in Table 4-1.

$$\lambda = \frac{c}{f} \quad , \quad d = 4t \quad , \quad L = \frac{\lambda}{2}$$

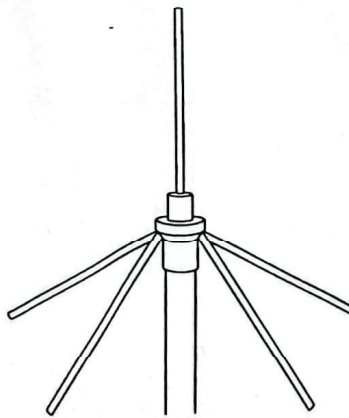


Figure 4-12 Drooping Antenna

Table 4-1 Length of Drooping Antenna [cm]

	Theoretical Antenna Length	Actual Antenna Length
λ		
L		
d		
t		

5. As Figure 4-13, the distance between antennas is isolated as $r=1.5\text{m}$. Make the center of each transmitting-receiving antenna to be in a straight line at a same height.

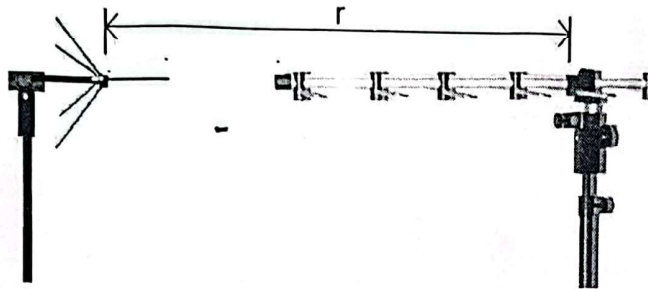


Figure 4-13 Distance between Antennas r

6. Set in condition of the power of a main controller as below.
- Power OFF
 - 500MHz Oscillation Switch off
 - 2 GHz Oscillation Switch off
 - 10 GHz Oscillation Switch off
 - Modulation Switch (Mod) off
7. After set a switch in main controller as below, execute the Antenna Trainer Program.
- Power ON
 - 500MHz Oscillation Switch on
 - 2 GHz Oscillation Switch off
 - 10 GHz Oscillation Switch off
 - Modulation Switch (Mod) on

8. The program window is executed, make the data acquisition in Pop Up Menu to be executed. Here after select Antenna, select E-Plane as Plane, and control the signal attenuation to be displayed 0 in a display for an antenna gain. The control is completed, start the data acquisition.
9. Data acquisition is finished, save the taken radial pattern in file. To confirm the radial pattern exactly, use a data box. Set MAP(Max. Signal Position) to 0° , can see a radial pattern of this antenna exactly.
10. For the 500MHz Yagi antenna used for transmitting get a vertical polarization characteristic, loose a fixing screw on a polarization control plate and make the control plate to rotate as 90° .

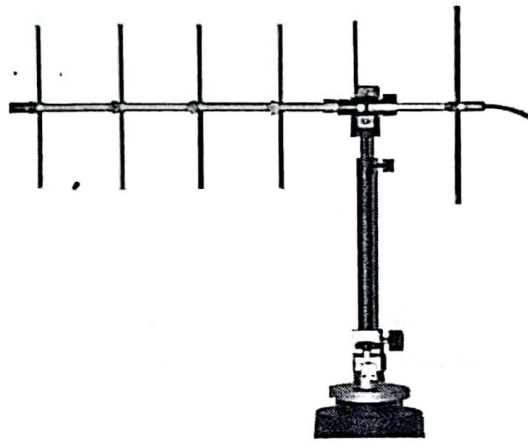


Figure 4-14 Setting of a Vertical Polarization Transmitting Antenna

11. For the drooping antenna used for receiving get a vertical polarization characteristic, set as Figure 4-15.

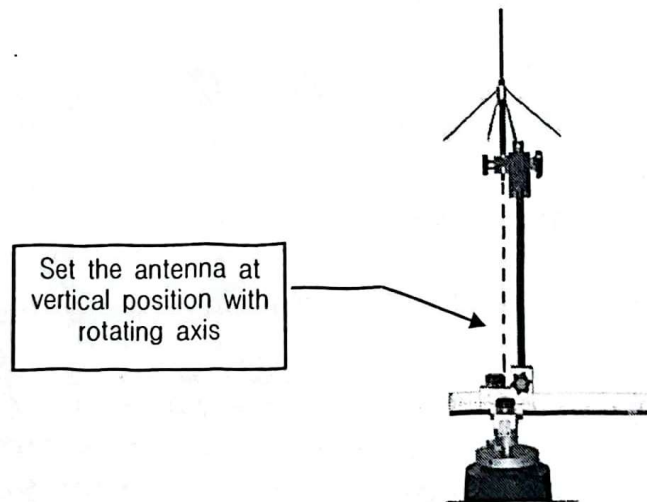


Figure 4-15 Setting of a Vertical Polarization Receiving Antenna

12. After click Antenna Initialize of Edit Menu of a software, acquire a data by clicking the Auto button. The data acquisition is completed, save the pattern in file after select H-Plane as Plane.
13. Keep a 500MHz Yagi antenna used for transmitting to get a vertical polarization characteristic, and set a drooping antenna to get an even polarization characteristic. In this time, for the center of a transmitting antenna taken a vertical polarization characteristic and a receiving antenna get an even polarization characteristic is to be in strait line, see Figure 4-16 to set.

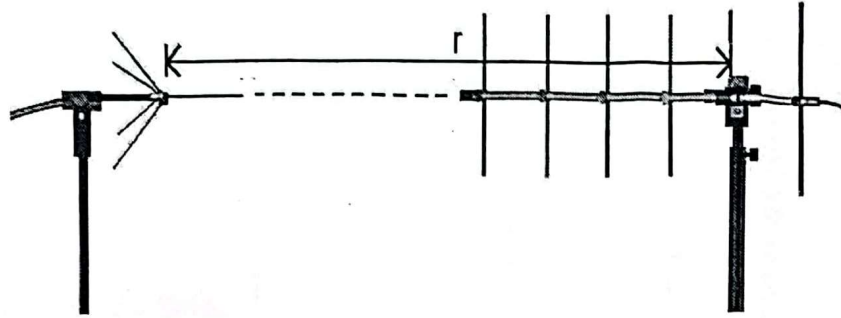


Figure 4-16 Vertical and Even Antenna

14. Click Antenna Initialize in Edit Menu of a software, and click Delete Screen in Edit Menu again, then acquire a data by clicking the Auto button.
15. Observe three radial pattern.
16. Set again each transmitting-receiving antenna to get a vertical polarization characteristic. The distance between antennas is $r=1.5\text{m}$. Make a surrounding condition to same as the first experiment. Take a E-Plane radial pattern of a drooping antenna and save in Dummy-Plane.

Theoretically except an effect caused by a power damage, this radial pattern must same as the first pattern. If the pattern is differ very much, have to do suitably by finding a point where a reflection can be occurred. After do suitably, continue the experiment and save a new pattern in E-plane.

Click the data indicating line in the Edit Menu of upper menu of a data analysis program. Two cursors(Blue, White) used for E-Plane and two cursors(Yellow, Green) used for H-Plane are appeared. Click the arrow mark at both side of a color displayer, the value in a displayer by moving the cursor of same color. These display the position angle of cursor and the receiving power as dB, The difference of an angle between cursors from a power on below two displayer. Record the values in below Table 4-2.

Table 4-2 Receiving Power for Angle

Angle	0°	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°
Receiving Power of E-Plane for Angle												
Receiving Power of H-Plane for Angle												

17. Take a receiving angle of max. power in E-Plane pattern and the receiving power in that time by using a cursor. Confirm if the measured value and the theoretic value are accorded by clicking the Maximum Single Level of E-Plane data window. Confirm if the taken value by using a cursor and record in below Table 4-3.

Table 4-3 Comparison of E-Plane with Theoretic Value

Taken Value by Using a Cursor	Receiving Power	
	Angle	
Value by Maximum Signal Level Icon	Receiving Power	
	Angle	

18. Take half angel of a main beam power in E-Plane pattern by using two cursors. Click the Half Power Beamwidth button.

Calculate a half power beam width of a drooping antenna in E-Plane by using below formula.

$$HPBW_E = | \theta_{HPWleft} - \theta_{HPWright} | = \underline{\hspace{2cm}}$$

19. Repeat the experiment of no. 18 in radial pattern of H-Plane.

$$HPBW_h = \underline{\hspace{2cm}}$$

20. Delete all cursors by clicking a data indicating line in Edit Menu. Compare the experimental values with a given values in an antenna software. If the calculated result isn't accord with these values, experiment and calculate again.
21. Save the data of E-Plane and H-Plane, and output the result.
22. Experiment from above no. 3 to no. 21 with taking the place of a ground plate(Large, Small) based on the experiment procedure. And compare this experimental result with above result, record.

▣ EXERCISE ▣

1. Where is the using place mainly for a drooping antenna, Why so?

2. What is the role of four lead placed in each corner of a drooping antenna?

3. What's the shape of a radial pattern for a drooping antenna?

4. The impedance and radial pattern according to a gradient angle of four lead placed on each corner of a drooping antenna. Take an impedance and radial pattern in case of the gradient angle of four lead is 30° , 45° , 60° , 90° .

5. What is the radial characteristic with a drooping antenna?
